

SOFTWARE ENGINEERING LAB3

Written Justification – Layered Architecture

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Selected Architecture: Layered Architecture:

The Layered Architecture is selected because the Personalized Meal Diet System has a clear separation between the Presentation, Business Logic, and Data layers. This makes the system easier to understand and maintain.

Reason 1 – Clear Separation of Responsibilities

The Presentation Layer handles user interaction, the Business Logic Layer manages profiles, meal plans, and deliveries, and the Data Layer stores meal and subscription data.

Reason 2 – Alignment with System Requirements

The architecture directly supports requirements such as personalized meal-plan generation, dietary constraints, dietitian review, and delivery tracking.

Security Advantage

The layered structure provides controlled access to business operations and stored subscriber data, allowing security mechanisms to be applied at appropriate layers.

Performance Benefit

The layers can be optimized independently to support the requirement of generating and displaying a weekly meal plan within 5 seconds under normal conditions.

Conclusion

Layered Architecture is the most suitable choice because it provides clear separation, requirement alignment, security, and performance benefits. Microservices may add unnecessary complexity, while Client-Server can create scalability bottlenecks.